Documented by:
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On behalf of:
True Interaction

LTB Sales Portal - System Development Progress

First Milestone

The LTB Sales Portal software is to provide users with permissions on four different layers in the system such as Group/User, Item, Finance and Operations. To meet the client's needs and requirements we have decided to associate users to a group, which will grant access to functions assigned to this group by a super user.

Milestone Goals:
There are two sections that we are achieving in this milestone:

1. The first section is understanding the company's goal and its data as a whole.
 -The system is to provide the company with the flexibility to grant their employees access to functions according to their capabilities. In this manner, the system can be more productive because it is both top down and bottom up; where upper and middle management levels can work together to finalize a plan.

2. The second section is designing the system architecture and its data structure.
 -The SUPER USER is the parent class that will contain most functions of the system. A super user can create groups and users with either basic or full permissions, where a user with full permissions could act as a super user if this has permissions to all layers such as Group/User, Finance, Item and Operations. All functions in the parent class are automatically inherited, and users can access these functions based on the permissions of their group. A USER class can represent an employee such as a Sales or Finance Person, Customer, Owner, among others. Similarly, a GROUP object represents a group of users such as a Sales or Finance Department, Company, Customers group and so on. Lastly, users with special permissions, such as division heads, are added to a special group with higher permissions.

Project Progress:

1. Company's Goals
The annual plan anticipates the coming year's activities that will contribute to the accomplishment of the long-term and short-term goals outlined in the company's strategic plan. This plan expresses management's operating and financial plans for a specified period (usually a year) and it reflects the impact of both, operating and financing decisions. The operating decisions deal with the use of scarce resources and the financing decisions deal with how to obtain the funds to acquire those resources.

2. Understanding Data
 a. <u>Sales Plan:</u>
  1) The sales budget projects the volume of sales for the coming period in both units and dollars and it is used as the basis for the preparation of all other budgets.
   - Preparation of the sales budget for a period starts with the system forecasted sales level, production capacity, and long-term and short-term objectives.
   - A sales budget is the cornerstone of budget preparation because the system can complete the plan for other activities only after it identifies the expected sales level.

- The system cannot complete the production schedule without knowing the number of units it must produce, and the number of units to be produced can be ascertained only after the system knows the number of units budgeted to be sold for the period.
- Once the number of units to be sold and produced has been determined, the units of materials to be purchased, the number of employees needed for the operation, and the required factory overheads can be determined.
- Expected selling and administrative expenses also are determined by the desired sales level.

2) The role of sales budget in the development of the annual profit plan:
   - Preparation of a sales budget is the first step in the comprehensive budget process.
   - Once sales have been determined, related budgets can be prepared, such as the production budget, expense budgets, and pro-forma financial statements.
3) Components of Sales Planning:
   - Production Budget
     - Direct materials budget
     - Direct labor budget
     - Manufacturing overhead budget
     - Ending finished goods inventory budget
     - Cost of goods sold budget
     - Pro-Forma Income Statement
   - Merchandising (Nonmanufacturing): (Selling, General & Administration) budget -> (The value chain activities)
     - Research and development budget
     - Design budget
     - Marketing budget
     - Distribution budget
     - Customer service budget
     - Administrative budget
     - Pro-Forma Income Statement

b. <u>Production Budget:</u>

A production budget is a plan for acquiring and combining the resources needed to carry out the manufacturing operations that allow the system to satisfy the sales goals and have the desired amount of inventory at the end of the budget period.

1) The total number of units to be produced depends on:
   - The budgeted sales
   - The desired amount of finished goods ending inventory, and
   - The units of finished goods beginning inventory
2) Future Period Budget:
   - The system does not know what the actual sales and production volumes will be.
   - All revenues and costs in the master budget are based on forecasted volumes.
   - Once the system knows the expected sales, production is estimated.
   - The production budget is based on assumptions appearing in the sales budget.
3) The master budget is a static plan because it is developed for one specific sales level:
   - Variance reports are prepared by comparing actual results to budgeted results.
   - Causes for variances are reported as well.

   Note: *Most common cause is that the actual sales volume was different from planned volume.*
4) The system must balance opposite goals to selecting a desired ending inventory for a period:
   - If the system has insufficient inventory, then sales are affected. And
   - If the system has an excessive amount of inventory, then it becomes costly.
   - The system must determine how quickly it can increase production if demand warrants and
   - Set its desired production and inventory levels accordingly.

5) The system needs to have the ability to coordinate sales needs and production activities to have a plan that allows the company to identify mismatches between capacity and output. As a result, these are some factors that should be considered when preparing a production budget:
   - Managers review the budget to ascertain that the company can:
     - Attain the budgeted level of production with the facilities available
     - Keeping in mind all activities scheduled for that period.
   - If the production level exceeds the maximum capacity available:
     - Management can either revise the budgeted sales level or
     - Find alternatives to satisfy the demand.
   - If the available capacity exceeds the budgeted production level:
     - Management might want to find alternative uses for the idle capacity
     - Schedule other activities such as
       - Preventive maintenance and
       - Trial runs of new production processes
6) The production plan budgets are interrelated since a change in one may require a change in another. This is the relationship between the direct materials, labor and other production budgets:
   - As production changes, the amount of labor the material changes.
   - As the amount of labor changes, there may be changes in indirect materials and labor. *These are used in manufacturing but their costs are not directly traceable to any particular product.*
   - As these items change, there may also be a change required in the overhead budget.
   - Because of how individual budgets are connected, a change in one is very likely to affect another.
7) Production Planning Components:
   - Direct materials budget
   - Direct labor budget
   - Manufacturing overhead budget
   - Ending finished goods inventory budget
   - Cost of goods sold budge
8) From calculating the production budget, in units, to Pro-Forma:
   - The production budget determines the units to be produced by adding the desired ending inventory to the units estimated to be sold and then subtracting the beginning inventory.
   - The budgets for direct materials, labor and factory overhead for the coming period are based on the required production from the production schedule.
   - The cost of goods sold budget may be prepared upon completion of the above budgets.
   - Once the level of activity has been determined for sales and production, the selling and administrative expenses budget can be prepared.
   - Completing these individual budgets allows the preparation of the Pro-Forma income statement
   - The cash budget shows the estimated flow of cash and the timing of receipts and disbursements based on projected revenues, the production schedule, and the estimated expenses.
   - The capital expenditures budget is a plan for the timing of acquisitions of buildings, equipment, and other significant assets.

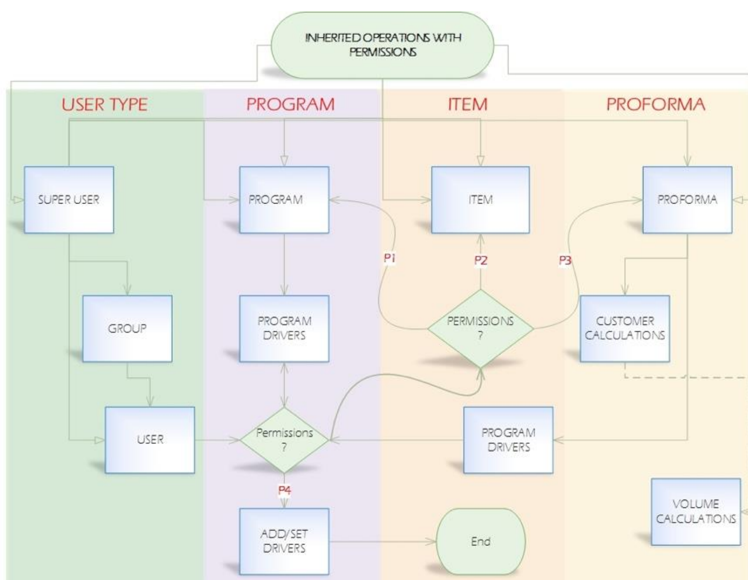
c. <u>Finance Plan</u>

A financial budget focuses on how operations and planned capital outlays affect cash. Financial plan is a set of budgets that identify sources of funds for the budgeted operation and the uses for these funds during a period to carry out the budget activities. In the financial budget, the emphasis is on obtaining the funds needed to purchase operating assets. In other words, financial planning is a continuous process that flows with strategic decision making. The Operating Plan and the Financial Plan will both support the Strategic Plan. The best place to start in preparing a budget is with sales since this is a driving force behind much of our financial activity. However, the system has to take into account numerous factors before finalizing budgets.

1) The system should allow modifications if something changes. The following impacts planning:
   - Life cycle of the business
   - Financial conditions of the business
   - General economic conditions
   - Competitive situation
   - Technology trends
   - Availability of resources
2) The system should allow both upper and middle level management work to finalize a budget. Challenges within financial planning to make planning value-added:
   - Clear channels of communication
   - Support from upper-level management
   - Participation from various personnel
   - Predictive characteristics
   - Budgeting should not strive for accuracy, but to support the decision making process
   - Make the organization achieve its strategic goals and objectives
3) Components of a financial plan - The financial budget is made up of the capital expenditures budget, the cash budget, the budgeted balance sheet, and the budgeted statement of cash flows:
   - Capital Budget (completed before operating budget is begun)
   - Cash Budget
   - Pro Forma Balance Sheet
   - Pro Forma Statement of Cash Flows
   - The master budget is finalized only after several rounds of discussions between top management and managers responsible for various business functions of the value chain.

3. System Architecture

a. <u>User Type</u>



In general, there are three objects used to identify customers, employees and company. The SUPER USER is the main class that contains all main functions of the system. This can create groups and users with either basic or full permissions, a user with full permissions could act as a SUPER USER if this contains full permissions for User Type(Employer), Program, Item, and Proforma. All functions in SUPER USER class are automatically inherited by groups and users, whom can access these based on the permissions set by a SUPER USER or a user with Employer permissions. An USER class can represent an employ, such as a Sales Person, Finance Person, Owner, among others. Similarly, a GROUP class is used to represent a group of users, such as a Sales Department, Finance Department, Company, Customers group, and so on. A user with Employer Full Permissions is able to add group(s) and users, where as a user with Basic Permissions is not.

b. <u>Sales Permissions</u>

1) Sales permissions are those that an employee has to sell item(s) to a customer. There are two types of permissions, Full and Basic. Full permissions provide the employee with full control, except for

validation, of a proforma for a specific department. Basic permissions provide the employee with enough access to create a proforma, submit and use this for selling purpose.

- *Sales Full Permissions:* In this case, only a division administrator will have Full Permissions to a Proforma. This means that the administrator of a division will be able to accept and reject proformas of items in that division. Full Permissions could also include Employer Full Permissions.
- *Sales Basic Permissions:* All sales people will have basic permissions, which can easily be modified by a SUPER USER or a user with the right permissions. These permissions will allow a sales person to create a proforma to sell an item.

c. <u>Finance Permissions</u>

1) Finance permissions are those that an employee has to create a program and set values that will drive the pricing of an item sold by a sales person. Similarly, Finance has Full and Basic permissions, where Full permissions provided Employees with Employer Full Permissions and Basic permission do no.

- *Finance Full Permissions:* finance managers will have full permissions to add, edit or remove an employee.
- *Finance Basic Permissions:* All finance employees, even managers, have Basic permissions to allow them create, edit, and delete programs from the profile of a customer, and to validate proformas.

d. <u>Program</u>

1) Programs are created by users with Finance permissions to set the global and default values that will drive the pricing of items for customers. A customer can have multiple programs since this varies based on the divisions that the items are from. Therefore, a user with finance permissions can create a program with a division and another program with another division generating in this way different pricing drivers for the same customer.

- *Add/Set Drivers:* It is yet to determine what group does the user that add drivers belong to. Nevertheless, a SUPER USER or user with Employer permissions is in charge of providing the right user access to this function to add drivers.

e. <u>Item</u>

1) An item object is created for every product that a company has, and they can be accessed by users with permissions. There are two types of permissions for an item, these are the Full permissions and Basic permissions.

- *Item Full Permissions:* Only Sales people will have access to view and edit items properties as well as their status.
- *Item Basic Permissions:* Every employee will be able to have basic permissions for items, which allow them to create and view items. All USERS, including customers, can view all items unless specified otherwise.

f. <u>Proforma</u>

1) A proforma is created to provide customers with a projection of the calculated cost and price of an item as well as the customer and the company margins. Permissions for a proformas can also be Full and Basic.

- *Proforma Full Permissions:* Only an administrator of a division will have full permissions of a Proforma, which allow the administrator to accept or reject the items pertaining to their division.
- *Proforma Basic Permissions:* All Sales People will inherit Basic permissions to create, edit and delete a proforma.

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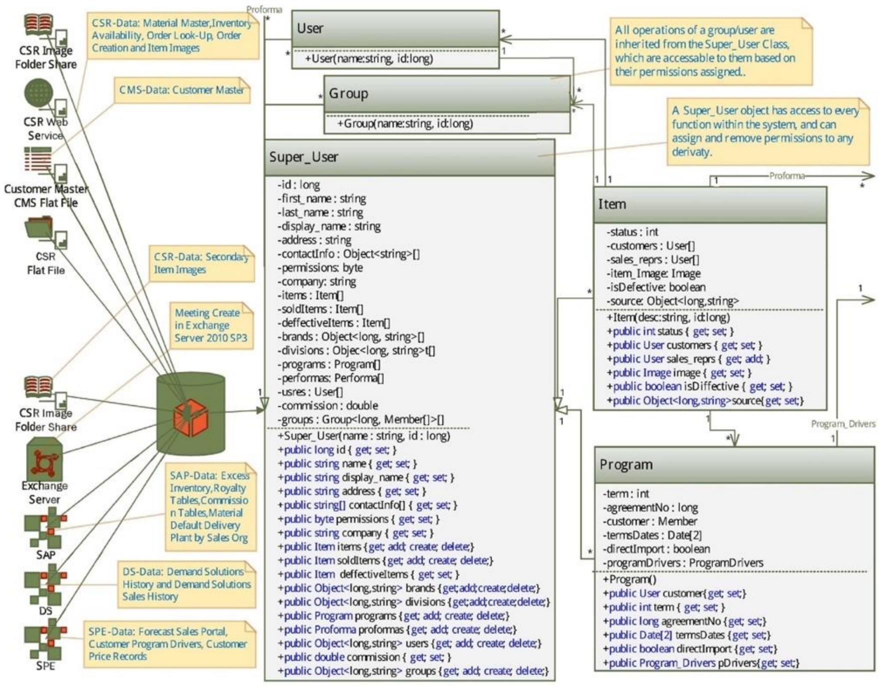

4. Data Structure

a. Super User: Main class that contains most of the main functions of the system. These functions address every action that takes place in all groups, especially those that pertain to finance, sales and customers. It is also responsible to save, update and access information, provided by the user, from/to the controllers/view.

b. Groups: Class used to relate a list of users to a unique user id that represents a unique group. This group inherits all functions in the Super User, which will be accessible according to the permissions set by a SUPER USER or a user with Employer permissions. Both users have permissions to create new users and, or, groups and provide these with permissions.

c. User: Class used to relate an employee, or customer, to a unique user id. The group class id will determine what type of user this is, such as finance person, sales person, customer, among others. The permissions of this user will also be set by a super user or a user with Employer permissions.

d. Program: Class used to set specific values that will drive the pricing of an item (product) for a specific customer. These drivers can only be set by a user with a finance group id.

e. Item: Class used to contain all information of an item, or product, and keep track of its status. Every user in this model is directly, or indirectly, related to an item. The selling process of an item involves a customer, sales person, division administrator and finance person in order to take place. The information of these, along with the properties of the item, are tracked and constantly updated.

f. Program Drivers: This class works along with the Program class to take care of the main drivers as well as accruals and non-accruals default drivers. This class can only be accessed by a user with Finance permissions.

g. Set Drivers: This class works along with the Program Drivers class to set the default accrual and non-accrual drivers of a specific customer Same as Program Drivers a person with Finance permissions can only set these drivers.

h. Driver: This class is used to represent a unique driver for a specific customer. A list of this class form a list of values that drive the pricing of an item for a specific customer. Since this class is directly related to the Program, Program_Drivers, and Set_Drivers classes, these can only be set by a user with Finance permissions.

i. Proforma: A class used to calculate the cost of an item, its selling price, and the profit generated by this. This class makes use of the Program and User(customer) and Item objects to make the necessary calculations. This class can only be accessed by a user with Sales permissions, accepted by a user with higher Sales permissions, and validated by a user with Finance permissions.

j. Volume: This class is used to calculate the quantity, orders, among other properties of an item annually.

k. Customer: This class is used to calculate the annual invoice price, retail price and margin of a customer.

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Next Step:

Permissions Sample Approach

V =	View
C =	Create & View
E =	Edit & View
D =	Delete & View

- Functions: byte [] ⟹

V	C	E	D
0	1	2	4

Functions		V	C	E	D
User					
Group					
Program					
Proforma	Validation				
Item	Status				
Planning					
Collection					
Performance					

Permissions Table

			Proforma		Item					
User	**Group**	**Program**	**Proforma**	**Validation**	**Item**	**Status**	**Planning**	**Collection**	**Performance**	**Total**
7	7	7	7	7	7	7	7	7	7	70
6	6	6	6	6	6	6	6	6	6	60
5	5	5	5	5	5	5	5	5	5	50
4	4	4	4	4	4	4	4	4	4	40
3	3	3	3	3	3	3	3	3	3	30
2	2	2	2	2	2	2	2	2	2	20
1	1	1	1	1	1	1	1	1	1	10
0	0	0	0	0	0	0	0	0	0	0

User Type		**Employer**		**Finance**	**Sales**					**Item**		
	Id	**User**	**Group**	**Program**	**Item**	**Item**	**Planning**	**Collection**	**Performance**	**Item**	**Status**	**Total**
Super User	0001	7	7	7	7	7	7	7	7	7	7	**70**
		14		**14**		**28**				14		
Sales Admin	0002	7	2	0	1	7	3	3	0	3	2	**28**
		9		**1**		**13**				**5**		
Sales Agent	0003	0	0	0	0	7	3	3	0	3	2	**18**
		0		**0**		**13**				**5**		
Finance Agent	0004	0	0	7	6	0	0	0	0	1	0	**14**
		0		**13**		**0**				**1**		
Finance Admin	0005	7	2	7	6	0	0	0	0	1	0	**23**
		9		**13**		**0**				**1**		
Customer	0006	2	0	0	0	0	0	0	0	0	2	**4**
		0		**0**		**0**				**1**		

User Validation – Database Table Sample

	0	**1**	**2**	**4**
0	0	1	2	4
1	1	2	3	5
2	2	3	4	6

0	➔ View
1	➔ Create, View
2	➔ Edit, View
4	➔ Delete, View

3	➔ Reject, Accept, Validate
5	➔ Reject, Accept, Invalidate
6	➔ Validate, Invalidate
7	➔ Reject, Accept, Validate, Invalidate

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Permissions Combination

User Permissions Tables

Groups Table		
	User Id	**Groups**
-1	0001	10XX (Sales)
-2		11XX (Finance)
-3		12XX (Employer)
-4		13XX (Item)

Employer Table				
GroupId	**UserId**	**User**	**Group**	**Total**
12XX	0001	7	7	14

Finance Table				
GroupId	**UserId**	**Program**	**Proforma Validation**	**Total**
11XX	0001	7	7	14

Sales Table						
GroupId	**UserId**	**Proforma**	**Planning**	**Collections**	**Performance**	**Total**
10XX	0001	7	7	7	7	28

1. As per the outline of this project, the next phase of this project will focus on the analysis of the data provided by LTB Sales team to prepare and explore the data, and to create data models after evaluating the data provided including the LTB-Sales-Portal-Requested-Features.

2. The wire frames for the graphics user interface of the system have been designed, and can be accessed by clicking on Wires/Frames-for-User-Interface. The data structure of the system heavily depends on the tasks that the LTB Sales team needs, as well as the style they want this to be presented and performed. An illustration of the data structure models and methods needed can be seen in the User Interface Data Structure document of this project.

Reference Files:

System and Data Structure Design **Week 0**
-- Sources

- Understanding the Company's Goal — LTB-Sales-Portal-Requested-Features
- Understanding the Data — Understanding-Data-Directory
- Design System Architecture — Data-Schema/PDFs/Model-Integration-System
- Design Data Structure — Wires/Frames-for-User-Interface